ORIGINAL ARTICLE

# Mental Illness in the Biological and Adoptive Relatives of Schizophrenic Adoptees

## *Replication of the Copenhagen Study in the Rest of Denmark*

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**Background:** Our previous investigation of the prevalence of mental illness among the biological and adoptive relatives of schizophrenic adoptees in Copenhagen, Denmark, showed a significant concentration of chronic schizophrenia (5.6%) and what Bleuler called "latent schizophrenia" (14.8%) in the biological relatives of chronic schizophrenic adoptees, indicating the operation of heritable factors in the liability for schizophrenic illness.

**Methods:** We now report the results of a replication of that study in the rest of Denmark (the "Provincial Sample").

**Results:** In this sample, the corresponding prevalences were 4.7% and 8.2%. In the combined "National Sample" of adoptees with chronic schizophrenia, that disorder was found exclusively in their biological relatives and its prevalence overall was 10 times greater than that in the biological relatives of controls.

**Conclusions:** This study and its confirmation of previous results in the Copenhagen Study speak for a syndrome that can be reliably recognized in which genetic factors play a significant etiologic role. These findings provide important and necessary support for the assumption often made in family studies: observed familial clustering in schizophrenia is an expression of shared genetic factors.

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THE TENDENCY displayed by several of the psychiatric syndromes to occur in other members of the patient's family has been recognized since these syndromes were first described and has been used variously to support genetic or environmental inferences of origin. To what extent the symptoms, syndromes, and diagnoses in the natural relatives of patients reared together are affected by genetic or environmental factors, enhanced in frequency by ascertainment and subjective bias, and molded or confounded by the relatives' close association with the patient, is difficult to specify, leaving conclusions regarding origin indeterminate and controversial. Where adoption has separated the genetic and familial environmental influences, they can more readily be differentiated and evaluated. Mental disorders, when they occur in the adoptee and some of the biological relatives, have developed independently and can be ascertained and diagnosed without interference from relatives' behavior or diagnosis. An increased prevalence of the disorder or one of the characteristics that constitute it in the biological or adoptive relatives when compared with suitable matched control groups would be compatible with the operation in them of genetic or family-associated environmental factors, respectively.

In 1963 a compilation of adoptees was begun in Denmark designed to provide the basis for an examination of the prevalence of schizophrenia in the nuclear biological and adoptive families of individuals adopted at an early age who eventually developed schizophrenia.[1] To our knowledge, adoption strategies had not been applied previously to the problem of schizophrenia and it seemed likely that

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## METHODS

Completion of the adoption register for the remainder of Denmark (the Provincial Sample), selection of probands, gathering of clinical information, and analysis of results proceeded in a manner similar to that involved in the Copenhagen Sample,[1] and it began with a search of the records of the Department of Justice that documents every legal adoption. The compilation of the Provincial Sample, designed to complete the national sample, started with a chronological search of the department's records from 1924 through 1947, and included every adoption granted by the courts outside of the city and county of Copenhagen to others than biological relatives of the adoptee, for a total of 8944 adoptees. (The Copenhagen Sample, selected earlier, had comprised 5483 adoptees, for a combined total for Denmark of 14 427.) The name, sex, and birth date of the adoptee were recorded and a code number was assigned that served to identify the adoptee, rather than the name, through subsequent processing. The same demographic information with the inclusion of occupation was recorded for the biological and adoptive parents, usually the father, and, for adoptive parents, their income and financial status around the time of the adoption.

### STUDY OF THE PROVINCIAL SAMPLE BASED ON INSTITUTIONAL RECORDS

To identify the adoptees who had developed schizophrenia (the "index" adoptees), a search was made of the National Psychiatric Register and of individual hospital files for those adoptees who had ever been admitted to a mental hospital or psychiatric service of a general hospital. All of these facilities are part of the National Health Insurance system and report, with rare exceptions, to the Psychiatric Register, maintained by Eric Stromgren and Annalise Dupont at the Institute of Psychiatric Demography in Risskov, Denmark. The experience in Copenhagen indicated that about 95% of admissions to psychiatric facilities would thus be recovered.[1] The dates of hospital admission and discharge and each hospital discharge diagnosis were recorded. Where the information was compatible with the possibility that the subject's disorder was schizophrenia, an abstract of the institutional records was prepared in English by a Danish psychiatrist and edited to remove any family-related information.

Independent diagnoses were made by three raters in America (S.S.K., P.H.W., D. Rosenthal, PhD), who then conferred to formulate a consensus diagnosis of schizophrenic or nonschizophrenic disorder according to an agreed-on set of criteria for chronic, latent, and acute schizophrenia or for other diagnoses, based on the descriptions of Bleuler[9] and the *Diagnostic Manual* then in use in the United States (*DSM-II*[10]). This yielded 41 presumptive index adoptees with preliminary diagnoses of chronic, latent, or acute schizophrenia, based only on the Psychiatric Register and abstracts of hospital records.

To arrive at an estimate of the prevalence of schizophrenia and other diagnoses that would result from the application of our search and diagnostic techniques in the populations of biological and adoptive relatives of adoptees without mental illness in this sample, presumptive controls were chosen who had no record of admission to a psychiatric facility.[1] From a group of four or five such adoptees who approximated each index adoptee in age, sex, and socioeconomic class of the adoptive family, one was selected who most closely matched the index proband in time spent with a biological parent and in placement before transfer to adoptive parents, as well as the time of that transfer. Time spent with biological parents was generally short, with a mean of 5 weeks and a median of 1 week; index and control probands did not differ significantly in time spent with biological parents before placement. Since later interviews indicated that individuals with no history of psychiatric hospitalization were not necessarily free of current or past mental illness, the final controls comprised only those who were interviewed and found to be free of affective or schizophrenic spectrum disorders, current or past. Screening control probands for diagnoses other than those by which index probands are selected introduces a possible bias toward the observation of an excess of screened disorders among the biological relatives of the unscreened index probands, which may be incorrectly interpreted as evidence for coaggregation among disorders.[11] In the present study, eliminating control probands with diagnoses other than chronic schizophrenia introduces a potential bias toward observation of coaggregation of these disorders among the relatives of index probands with chronic schizophrenia who were not screened for the presence of latent schizophrenia or nonbipolar affective disorder. However, screening control probands allows more accurate estimation of the magnitude of risk for chronic schizophrenia in the biological relatives of controls.

Identification of relatives proceeded through the adoption and population registers as was the case with the Copenhagen Study. The adoption records identified the adoptive parents and the biological mother and putative biological father. Confidence in the latter information was enhanced by the requirement that the mother name the father and that he contribute to the costs of maternity care and the adoption process. In one instance the biological mother named two putative fathers and it was decided that the more conservative choice, made blindly, was to include both and their offspring, if any, in the subsequent analyses.

Offspring of the biological and adoptive parents were then identified through the Folkeregister, a remarkably comprehensive population register, which lists for each resident the present and all previous home addresses and members of the household, thus permitting the identification of siblings and half-siblings of the proband, both biological and adoptive, and often, the other biological parent in the case of half-siblings. The dates of death or emigration of any of the relatives were also noted.

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## DIAGNOSES IN RELATIVES BASED ON ABSTRACTS OF HOSPITAL RECORDS

The Psychiatric Register was then searched for admission of any relative to a psychiatric facility, and an English abstract was prepared from the institutional record and edited to remove any information regarding relationship to an index or control proband. Military or police records that had been searched in the Copenhagen Study had not yielded sufficient new information to warrant their continued use. The edited abstracts were read by three raters (S.S.K., P.H.W., and D. Rosenthal, PhD) in the United States, who made blind independent diagnoses, followed by consensus diagnoses. Agreement among raters' independent diaqnoses of schizophrenia, latent schizophrenia, acute schizophrenia, or other diagnosis in probands and relatives ranged from $\kappa$=0.71 to $\kappa$=0.77 by the $\kappa$ statistic of Cohen.[12] These were the basis of a preliminary report on the Provincial Sample, based only on institutional records, which appeared in 1978.[13]

## THE PSYCHIATRIC INTERVIEWS

Experience with the Copenhagen Sample indicated the desirability of augmenting the clinical information available on probands and relatives in the sample through careful follow-up and extensive personal interviews.[2] Nearly 90% of the relatives alive and known to be residing in Denmark or a neighboring Scandinavian country agreed to participate in a personal interview by a physician. Initial contact was through a letter to each living relative residing in Denmark, Norway, or Sweden, which if not answered was followed by a telephone call or visit by one of us (B.J.). The detailed interview was usually conducted in the subject's home and lasted from 1 to sometimes more than 3 hours. In the 6% to 9% of available relatives who refused to be interviewed, the circumstances surrounding the refusal and information from the subject prior to refusal were reported. In one instance the information was sufficient for the raters to make a blind diagnosis that was included in the analyses. All of the control adoptees and those index adoptees who were not in hospital were included, without being identified as such, among those asked to participate in an interview, and a large majority did so. The distribution of subjects who were interviewed, who refused to participate, who died prior to the study, or who were unavailable for follow-up is presented in **Table 1**.

Refusals were distributed nonsystematically among groups, but a greater proportion of adoptive relatives (both index and control) had died, reflecting older adoptive and younger biological parents. The smaller number of adoptive relatives compared with biological relatives also reflects the smaller size of adopting families. These observations duplicate those found in the Copenhagen Study.[2]

The structured interviews and hospital record summaries were more comprehensive than the Schedule for Affective Disorders and Schizophrenia–Lifetime version (SADS-L),[14] which had become available by that time, extending to 40 or more pages with numerous checklists and extensive narrative elaborations, designed to elicit information on a complete range of medical, psychological, and psychiatric symptoms and manifestations, particularly those that had been described in the literature for schizophrenia and schizophrenialike syndromes. The questions covered the major aspects of the life experience: sociological, interpersonal, educational, occupational, and marital; peer relationships; medical background; and a careful mental status examination. The observations were recorded in English soon after the interview and transcribed. The standard Schedule for Affective Disorders and Schizophrenia–Lifetime version questionnaire (in Danish) was also completed as were various notations of cognitive ability, verbal activity, and social functioning.

The interviews were conducted by one of the authors (L.J. or B.F.), psychiatrists trained in Copenhagen by interviewing patients on the ward to achieve reliable ratings, and who periodically met to ensure that reliability was maintained. The interviewers, when the interview began, did not know whether the subject was a proband or relative, index or control, biological or adoptive, or what relationship to a proband, and did not raise the issues of adoption or of mental illness in the subject's family. If the subject gave such unsolicited information, however, the semistructured interview required that it be noted in a particular place on the interview form. This, as well as other potentially biasing information, was deleted from copies of the interviews that were used for diagnosis. For every index proband, in addition to the original abstract of institutional records, more exhaustive summaries of these records were prepared on forms similar to the interview forms and were edited in the same manner. These were recognized as those of presumptive index probands, although which proband and the relationship to other subjects remained unknown. These as well as the edited interview transcripts were then read by

schizophrenic adoptees who shared their genetic endowment with their biological relatives but their family environment with others would make it possible to disentangle the genetic from the family-related environmental influences. The goal was a maximal national sample in which ascertainment of illness was based on exhaustive surveys of total population groups with the expectation that such a sample would provide an adequate number of schizophrenic probands with a minimum of ascertainment bias. If, in addition, diagnoses in probands and relatives were made without knowledge of their relationship, subjective bias would also be minimized.

Demographically matched controls who had no history of psychiatric hospitalization and appeared free of significant mental illness on interview were selected from the adoption register to arrive at an estimate of the prevalence of schizophrenia and other diagnoses that would result from the application of the same search and diagnostic techniques in the relatives of adoptees without mental illness in this sample. The probands were adults who two of the American authors (S.S.K. and P.H.W.), who made independent global diagnoses. The raters' independent diagnoses of the interviews agreed moderately well (κ=0.68) for the diagnoses of chronic schizophrenia, latent schizophrenia, acute schizophrenia, or other and no diagnosis. Following independent diagnoses, the raters conferred and made a consensus interview diagnosis for each subject.

A final comprehensive diagnosis was arrived at for probands and relatives, based on all of the consensus diagnoses made from abstracts and summaries of institutional records, refusal reports, and psychiatric interviews conducted without knowledge of the identity or relationship of any subject to another. In previous reports we have presented results based on hospital records and psychiatric interviews separately; in the present study we have based our diagnoses on a final comprehensive diagnosis for each individual.

## DIAGNOSTIC CRITERIA

Diagnoses, as in the case of the Copenhagen Study, represented global judgments based on symptom clusters described by Kraepelin[15] for dementia praecox and by Bleuler[9] for schizophrenia, less clearly demarcated by him for latent schizophrenia, and on the succinct encapsulations of *DSM-II*[10] for these and cognate disorders. Certain minor modifications evolved in our criteria or in their use after the earlier studies. These included a decreased utilization of the *DSM-II* diagnoses "acute schizophrenia" or "pseudoneurotic schizophrenia," probably because of their infrequency and lack of association with schizophrenia in the Copenhagen Study. There was also the recognition that the distinction between chronic and "latent" schizophrenia was demonstrable and significant for these diagnoses, whereas that between "definite" and "uncertain" or "probable" was not. Since classical schizophrenia and latent schizophrenia were to become the only forms of schizophrenic illness to be identified as such, there was no longer a need to characterize the former as "chronic." The diagnosis "schizoaffective" was rarely used since adherence to the classic descriptions of schizophrenia and manic-depressive disorder diminished its applicability. This was borne out by the independent *DSM-III* ratings on the Copenhagen interviews,[16] where the separate proband grouping of "schizoaffective, mainly schizophrenic" could not be differentiated from "schizophrenia" in the prevalence of schizophrenic illness in the results.

The negative features of schizophrenia and traits similar to them continued to play a major role in the characterization of schizophrenia and latent schizophrenia, and when they were present, affective disturbances were not exclusionary, in conformity with the following argument of Bleuler[9(p304)]: "All the phenomena of manic-depressive psychosis may also appear in our disease; the only decisive factor is the presence or absence of schizophrenic symptoms."

However, a description of the criteria we used in making the global diagnoses remains to be made and incorporated into an operational form. As a first step, the criteria regarded as important in our diagnosis of schizophrenia and their presence or absence in the probands for whom we made that diagnosis are presented in **Table 2**.

The diagnosis of latent schizophrenia, analogous but not identical with the *DSM-III*[17] diagnoses of schizotypal and paranoid personality disorders, was made following the *DSM-II*[10] description of latent schizophrenia, which emphasizes the presence of symptoms of schizophrenia in the absence of a psychotic episode, and following the description by Bleuler[9(pp238-239)] of latent schizophrenia:

> If one observes the relatives of our patients, one often finds in them peculiarities which are qualitatively identical with those of the patients themselves, so that the disease appears to be only a quantitative increase of the anomalies seen in the parents and siblings.
>
> There is also a latent schizophrenia, and I am convinced that this is the most frequent form, although admittedly these people hardly ever come for treatment. It is not necessary to give a detailed description of the various manifestations of latent schizophrenia. In this form, we can see in nuce all the symptoms and all the combination of symptoms that are present in the manifest type of the disease.

A diagnosis of latent schizophrenia was made where social withdrawal, affective flattening, changes in the form of thought and speech, or delusional (including paranoid) tendencies characteristic of chronic schizophrenia were present in the absence of frank delusions or other psychotic symptoms. The presence of characteristics important in making our diagnoses of latent schizophrenia among relatives of index and control probands is presented in **Table 3**. A diagnosis of schizoid personality in the present study most frequently reflected severe introversion, suspicious or referential symptoms in otherwise psychiatrically unremarkable individuals, and was not identical to the *DSM-III* diagnosis of schizoid personality.

**Table 1. Interview Participation by Provincial Sample Relatives***

| Relatives of Proband Type | No. (%) of Total: Unavailable for Follow-up | Died | Refused | Interviewed | Total No. | Interviewed, % of Available |
|---|---|---|---|---|---|---|
| Biological index | 7 (2.9) | 44 (18.0) | 22 (9.0) | 172 (70.2) | **245** | 88.6 |
| Adoptive index | 2 (2.0) | 52 (51.0) | 6 (5.9) | 42 (41.2) | **102** | 87.5 |
| Biological control | 7 (3.3) | 38 (17.5) | 17 (7.8) | 155 (71.4) | **217** | 90.1 |
| Adoptive control | 0 (0.0) | 46 (44.2) | 7 (6.7) | 51 (49.0) | **104** | 87.9 |

**Includes only relatives at least 15 years old.*

**Table 2. Course and Symptoms of Chronic Schizophrenia in Adoptee Probands**

| Good premorbid adjustment | Insidious onset | Schizoid features observed premorbidly | Age at first psychiatric hospitalization, y | Chronic course observed | Withdrawal from social interaction | Autistic behavior | Poverty of thought/speech | Loose associations | Flat affect | Suspicious/ideas of reference | Delusions | Auditory hallucinations | Other hallucinations | Family No. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | x | | 35 | x | | x | | | x | x | x | x | | 369 |
| | | x | 34 | x | x | x | | x | x | x | x | x | x | 420 |
| | | | 24 | | | x | x | | x | | | x | | 794 |
| | x | x | 25 | x | x | | | | x | x | x | x | | 241 |
| x | | | 26 | x | | | | x | | x | x | x | x | 368 |
| x | x | | 25 | x | x | x | x | | x | x | x | x | x | 830 |
| | x | | 33 | x | | x | | x | x | x | x | x | x | 936 |
| | x | x | 27 | x | x | x | | x | x | x | | | | 974 |
| x | x | x | 19 | x | x | | | | x | x | x | | | 078 |
| x | x | | 36 | x | x | | | | x | x | x | x | x | 258 |
| x | x | | 25 | x | | | | x | x | x | x | x | x | 991 |
| | x | | 26 | x | | | | x | x | x | x | x | | 055 |
| x | x | x | 20 | x | x | | | x | | x | | | | 828 |
| | x | | 25 | x | | | | x | | | x | | | 871 |
| x | x | x | 27 | x | x | x | | x | x | x | x | x | x | 975 |
| | x | x | 21 | x | x | x | | | x | x | x | x | x | 042 |
| x | x | x | 21 | x | x | | | | | x | x | x | x | 141 |
| | x | x | 17 | x | x | | | x | x | x | x | x | | 831 |
| x | x | x | 19 | x | x | x | x | x | x | x | x | x | | 069 |
| | x | x | 22 | x | x | x | x | x | x | x | x | x | x | 462 |
| | x | x | 11 | x | x | x | | | x | x | x | x | | 592 |
| x | x | | 12 | x | | | | x | | x | x | x | x | 913 |
| | x | x | 20 | x | x | x | | | x | x | x | x | x | 940 |
| | x | x | 14 | x | x | x | | x | x | x | x | x | x | 782 |
| x | x | x | 19 | x | x | x | x | x | x | x | x | x | x | 806 |
| | x | x | 17 | x | x | | | | x | x | x | x | x | 900 |
| x | x | | 21 | | | | x | x | x | x | x | | | 020 |
| | x | x | 24 | x | x | | | | x | | | | | 139 |
| x | x | | 20 | x | | | | x | | x | x | x | | 388 |

had been adopted early in life, separating, to the maximal extent possible, their genetic and environmental endowments and making possible independent evaluations of the significance of genetic and family-associated environmental factors in the development of their disorder. Results of the studies based on the adoptions granted in the city and county of Copenhagen, Denmark (the Copenhagen Sample), have been published previously using psychiatric information from institutional records[1] and from extensive personal interviews.[2]

Another strategy that the register of adoptions would serve was to be an examination of the risk for schizophrenia in the adopted-away offspring of a schizophrenic biological parent, compared with the risk in an adoptee whose biological parents were free of the disease.[3,4] There have been several studies by others of adopted-away offspring of schizophrenic mothers,[5-8] but to our knowledge, the Copenhagen Study of adoptees' families and the present replication in the rest of Denmark represent the only examinations of schizophrenia and other mental illness in the biological and adoptive relatives of schizophrenic adoptees that have been conducted.

## RESULTS

The interviews, detailed review of hospital records, and follow-up information on the initial group of potential proband adoptees made possible the selection of a final set of index probands who exemplified schizophrenia or latent schizophrenia, and a comparable set of control adoptees who had been interviewed by a psychiatrist and found to be free from serious or confounding mental illness (major affective disorder, chronic or latent schizophrenia, or schizoid personality). This decreased the number of control probands to 24 since a psychiatric interview was refused or unobtained in five patients and in 13 patients it identified a confounding illness among putative control probands who had never been hospitalized on a psychiatric service. The review identified 29 index probands with chronic schizophrenia and four index probands with latent schizophrenia among the original sample of potential schizophrenic probands on the basis of interviews, updated information, or new information.

At the same time that the interviews, hospital records, and follow-up information on the proband adoptees in the present (Provincial) sample were reviewed, the records of the Copenhagen sample of probands, augmented by the same additional information, were also reviewed in an effort to use all available information in identifying index probands with chronic (or latent) schizophrenia and control probands free from serious or confounding illness. Overall, this resulted in one additional proband with chronic schizophrenia and one fewer proband with latent schizophrenia (two probands diagnosed as having acute schizophrenia some 15 years earlier were now found to be better characterized by chronic schizophrenia, as was one proband with latent schizophrenia; for two probands, review indicated that the previous diagnosis of chronic schizophrenia should be revised to latent schizophrenia, as was the case for one proband previously diagnosed as having acute schizophrenia; four probands previously diagnosed as having latent schizophrenia were, on review, diagnosed as having nonschizophrenia spectrum diagnoses). Updated diagnoses were always made without knowledge that could identify the subjects or their relatives.

Study of the biological relatives of these schizophrenic and control probands permitted an examination of the hypothesis that there are one or more clinical syndromes genetically related to schizophrenia as well as the primary hypothesis that genetic factors operate in the familial transmission of schizophrenia. We removed from our analysis relatives who had died or emigrated before the age of 15 years to exclude relatives who had not entered the age of risk for schizophrenia.

**Figure 1** and **Figure 2** present the pedigrees of probands with chronic and latent schizophrenia in the Provincial Sample and the diagnoses of their biological and adoptive relatives. **Figure 3** similarly diagrams pedigrees and diagnoses in the families of control probands.

**Table 4** and **Table 5** summarize the results presented in Figures 1 and 3. Table 4 compares the prevalence of schizophrenia and latent schizophrenia (definite or probable) as well as schizoid and paranoid personality disorder in the biological and adoptive relatives of chronic schizophrenic index and control probands.

The figures and tables present the observed prevalence of schizophrenic illness in our sample, rather than a hypothetical lifetime prevalence. An age-corrected morbid risk can be estimated with the abridged Weinberg method (age of risk of 15 to 39 years).[18(pp38-42)] The age-corrected morbid risks for chronic schizophrenia and latent schizophrenia among the biological relatives of chronic schizophrenic index adoptees are 9.9% and 5.7%, respectively; the age-corrected risk for latent schizophrenia among the relatives of control probands is 2.9%.

Schizophrenia and latent schizophrenia are significantly more common in the biological relatives of adoptees with chronic schizophrenia than in the biological relatives of controls, and they are absent in adoptive relatives. The prevalence of schizoid and paranoid personality is not significantly different among groups.

Table 5 compares the prevalence of major affective disorder, other psychotic disorders, minor affective disorder, and all other diagnoses, and indicates that they occur at similar rates in the biological relatives of index and control probands. Among adoptive relatives, there are no significant differences between the adoptive relatives of schizophrenic or control adoptees. **Table 6** presents for comparison the results from the updated Copenhagen Sample, and **Table 7** presents the results of the combined Copenhagen and Provincial Samples as the National Sample.

**Table 3. Course and Symptoms of Latent Schizophrenia in Relatives of Probands With Chronic Schizophrenia and in Relatives of Control Probands**

| Referential delusive tendency | Social anxiety | Odd beliefs/magical thinking | Unusual perceptual experiences | Odd/eccentric/unkempt | No close friends/seclusive/withdrawn | Odd or digressive speech | Inappropriate or constricted affect | Suspicious/paranoid ideation | Anhedonic | Impaired reality testing | Age at psychiatric hospitalization, y | Family No. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | x | | | | | 420 |
| x | | | | | | | x | | | | | 794 |
| | | | | | x | | x | x | | | | 794 |
| | x | | | | x | | x | x | | | | 368 |
| | | | | | | | x | x | | | | 368 |
| | | | | | x | | x | | x | x | | 487 |
| | | x | | | | | | | | x | 28 | 830 |
| | | | | | | x | | | | | 66 | 936 |
| | x | x | | | x | | x | x | x | | | 974 |
| | | | | | x | x | x | x | | | | 258 |
| | | x | | | | | | | | | 46 | 837 |
| | | | | | x | | | x | | | | 871 |
| | | | | x | | | x | | | | | 497 |
| | | x | | | | | | | x | | | 940 |
| | | x | | | x | | x | x | x | | | 940 |
| x | | x | | | x | x | | x | | | 44 | 782 |
| | | | | x | x | x | x | | | | | 388 |

those in the control families, which were zero and three of 24 ($P$=.041 and $P$=.006), respectively.

## FAMILIES AFFECTED

It is possible to express the results in terms of biological families affected rather than relatives. Of the 29 biological families of probands with chronic schizophrenia, five had at least one schizophrenic member and 14 had one or more members with schizophrenia or latent schizophrenia. These frequencies are significantly different from those in the control families, which were zero and three of 24 ($P$=.041 and $P$=.006), respectively.

## FIRST- AND SECOND-DEGREE RELATIVES

A focus on the first-degree relatives of schizophrenic probands finds an excess of schizophrenia (six of 81) compared with those of controls (none of 60; $P$=.033). In disorders in which genetic influences appear to play a role, the distribution of illness between first- and second-degree biological relatives may be pertinent if the size of the samples involved is large enough to achieve statistical power. Since the abortion and placement policies in Denmark reduced the observed incidence of schizophrenia in parents of adoptees, our examination of the relative incidence of schizophrenia in first- and second-degree relatives focused on adoptees' full and half siblings. The Copenhagen Sample suffered from a paucity of full biological siblings that prevented a meaningful estimate

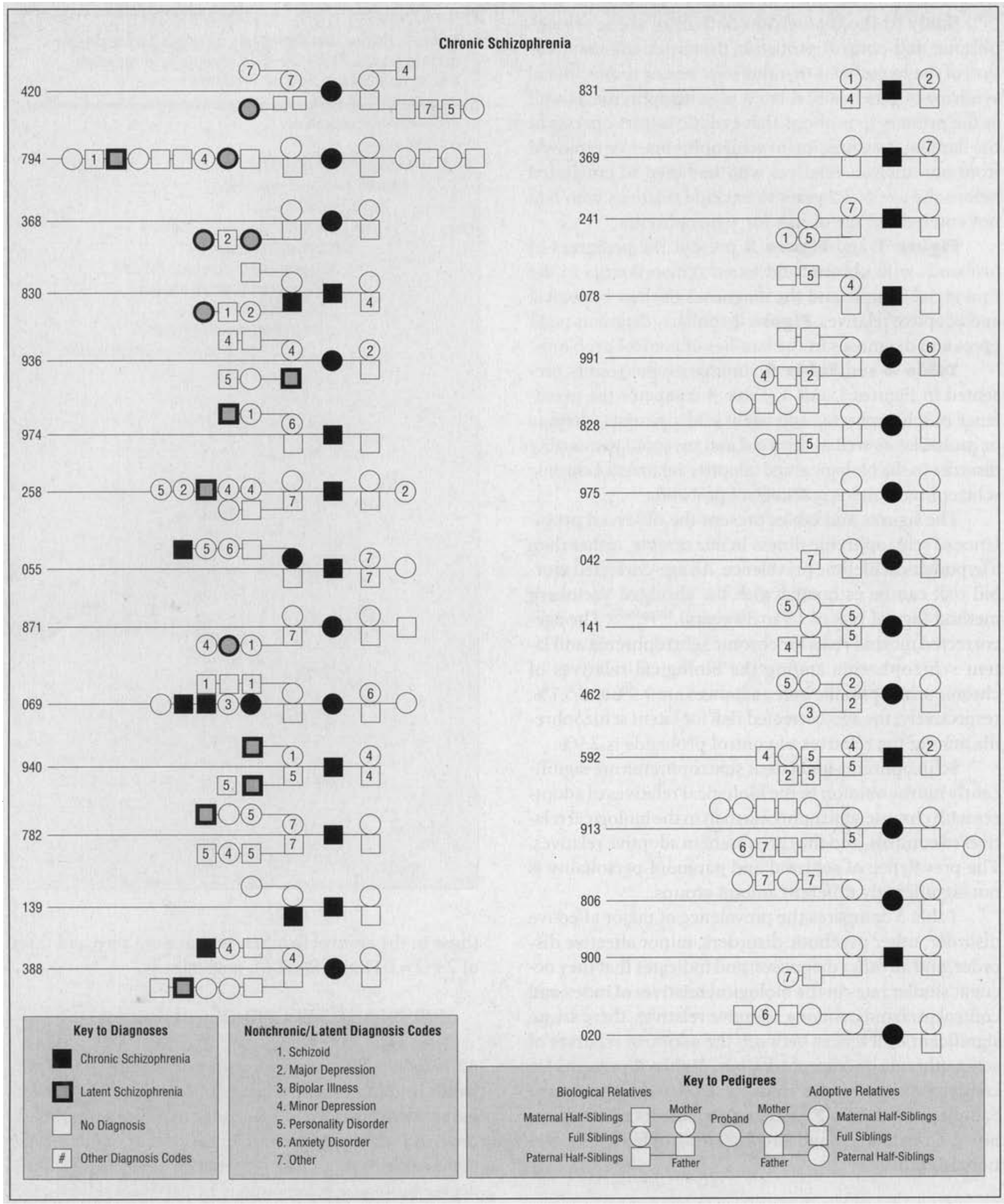


**Figure 1**. *Pedigrees of adoptees with chronic schizophrenia.*

of disease prevalence in first-degree relatives, but in the Provincial Sample there were a substantial number of full siblings (24 biological index siblings compared with three in the Copenhagen Sample), probably the result of differences in lifestyle from that of the large city. Schizophrenia was found in three (12.5%) of the 24 full siblings, nearly six times the rate in the half-siblings (two [2.2%] of 90) (one-tailed $P$=.062). (The less specific marginal syndrome [latent schizophrenia] was evenly distributed between full siblings and half-siblings [12.5%

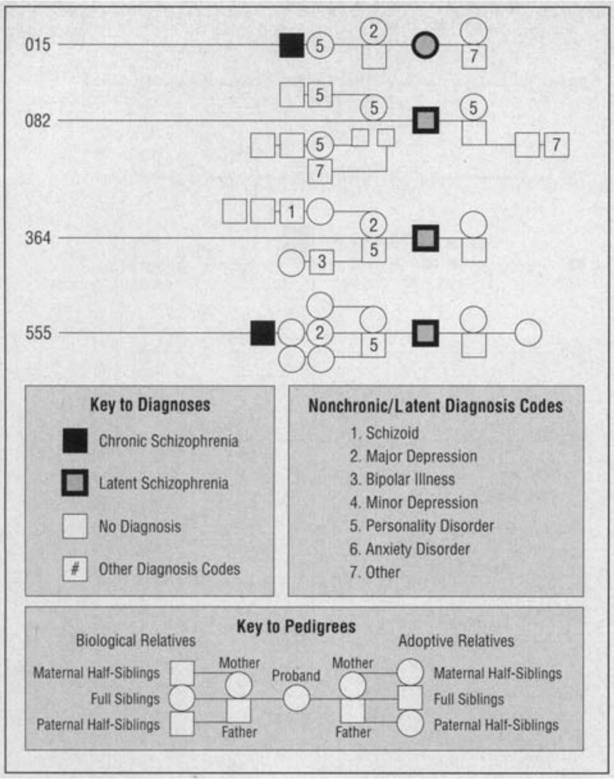


**Figure 2.** *Pedigrees of adoptees with latent schizophrenia*

vs 11.1%, respectively], suggesting a greater degree of heterogeneity and of nongenetic influence within that syndrome.)

## SPECIFICITY OF LATENT SCHIZOPHRENIA

The 8.2% prevalence of latent schizophrenia in the biological relatives of index probands was significantly higher than the 2.5% in the relatives of controls ($P$=.032), while paranoid or schizoid personality did not show significantly different prevalence, confirming similar findings in the Copenhagen Sample. Since a higher than expected prevalence of the features of schizotypal personality disorder (the *DSM-III* diagnosis derived from and comparable to our latent schizophrenia) was reported in the adolescent children of parents with affective disorder,[19] the occurrence of latent schizophrenia in the biological relatives of adoptees with affective disorder in this sample became of interest. Eight such adoptees had been identified through the psychiatric interviews (seven unipolar depressed and one bipolar manic) and among their 37 biological relatives a prevalence of 13.5% of latent schizophrenia was found, significantly higher ($P$=.015) than the prevalence in relatives of normal controls and not significantly different from the prevalence of that syndrome in the relatives of schizophrenics.

Another approach to investigating the relationship between chronic and latent schizophrenia focuses on whether there is an increased risk of chronic schizophrenia among the relatives of individuals with latent schizophrenia, as would be expected if they share common genetic liabilities.[20-22] In our limited sample of four probands with latent schizophrenia, we observed two instances of chronic schizophrenia among their 29 biological relatives (Figure 2), a prevalence of 6.9%; we did not observe latent schizophrenia among these relatives.

In sum, our present results find a substantial risk for latent schizophrenia among the biological relatives of probands with chronic schizophrenia, indicate that latent schizophrenia is not seen exclusively among the biological relatives of individuals with chronic schizophrenia, and suggest that latent schizophrenia may be associated with an increased liability to chronic schizophrenia among biological relatives. While the existence of a link between chronic schizophrenia and latent schizophrenia is supported by our results, further elucidation of the nature of the relationship between schizophrenia and latent schizophrenia is necessary.

## COMMENT

### EVIDENCE FOR THE OPERATION OF GENETIC FACTORS IN THE ETIOLOGY OF SCHIZOPHRENIA

The most striking finding in both this and the Copenhagen Study is the highly significant concentration of classical schizophrenia in the biological relatives of the adoptees with the same severe disorder: 4.7% and 5.6%, respectively. This is significantly higher than that found in the biological relatives of adoptees free of serious mental illness: 0% and 0.9%, respectively. For all of Denmark, schizophrenia was found in 14 (5.0%) of 279 biological relatives of schizophrenic adoptees compared with one (0.4%) of 234 biological relatives of controls ($P$=.0013). Moreover, schizophrenia was found almost exclusively in the biological relatives of index adoptees with classical schizophrenia in both this and the Copenhagen Study. Schizophrenia was found in 12.5% of the first-degree relatives (full siblings) compared with 2.2% of second-degree relatives in this study, also consistent with genetic transmission. The classical schizophrenia syndrome was found in five of 29 index families in this sample compared with none in 24 control families. All of these differences speak for a syndrome that can be reliably recognized, in which genetic factors play a significant etiological role. It should also be noted that this prevalence in the biological relatives of schizophrenic adoptees who were not reared by or with them, matching that in the relatives of naturally reared schizophrenics, provides important and needed support for the assumption usually made in family studies: that the familial clustering is an expression of genetic factors.

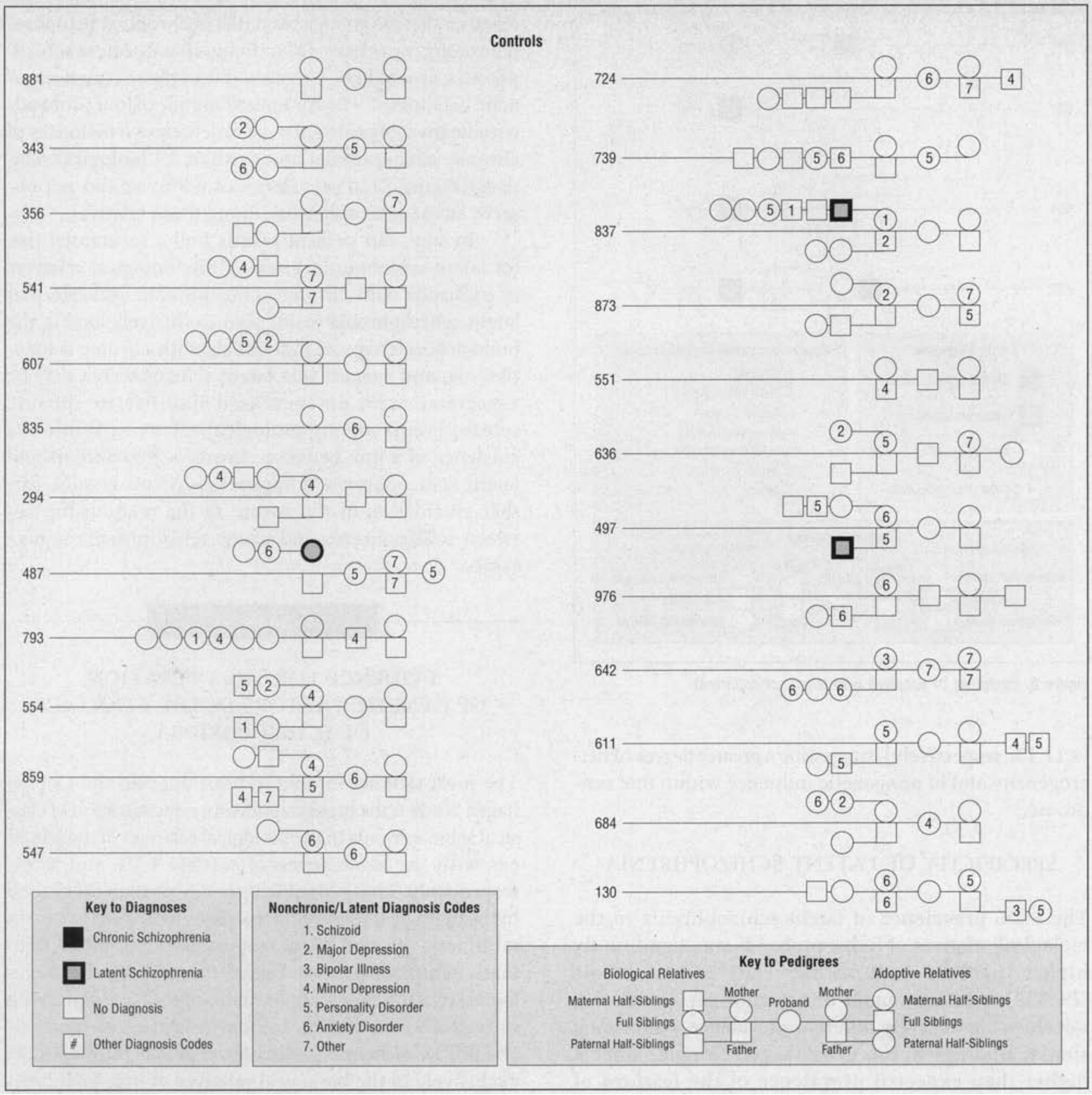


**Figure 3.** *Pedigrees of control adoptees.*

## LIMITS TO THE SCHIZOPHRENIA SPECTRUM

The marginal syndrome (latent schizophrenia) seen by Bleuler[9] in the relatives of schizophrenics is also found in the biological relatives of the schizophrenic adoptees in this and the Copenhagen sample: 8.2% and 14.8% compared with 2.5% and 0.9%, respectively, in the relatives of control adoptees, with a prevalence for the combined national sample of 10.8% vs 1.7% ($P$=.00002). Furthermore, chronic schizophrenia was found in two biological siblings of probands with latent schizophrenia in the Provincial Sample. While these results support the genetic relationship of this marginal syndrome to schizophrenia, its increased prevalence in the biological relatives of adoptees with affective disorders renders it less than completely specific to schizophrenia. It is possible that more precise diagnostic criteria could characterize a specifically schizophrenia-related syndrome.

Neither schizoid or paranoid personality nor any diagnosis outside of the schizophrenia spectrum appeared in significantly higher frequency in the relatives of index vs control adoptees. This suggests that increased risk for psychiatric illness in the biological relatives of schizophrenic individuals is limited to the diagnoses of schizophrenia and latent schizophrenia, rather than a risk for psychopathologic features in general or for a range of mild

**Table 4. Provincial Sample: Chronic Schizophrenia and Potentially Related Diagnoses in Relatives of Adoptees With Chronic Schizophrenia and in Relatives of Control Adoptees With No History of Major Mental Illness**

| Relative Groups | Paranoid, No. (%) | Schizoid, No. (%) | Latent Schizophrenia, No. (%)* | Chronic Schizophrenia, No. (%)† | Total No. of Relatives |
|---|---|---|---|---|---|
| Biological relatives of 29 adoptees with chronic schizophrenia | 3 (1.8) | 7 (4.1) | 14 (8.2) | 8 (4.7) | 171 |
| Biological relatives of 24 control adoptees | 1 (0.8) | 4 (3.3) | 3 (2.5) | 0 | 121 |
| Adoptive relatives of 29 adoptees with chronic schizophrenia | 0 | 0 | 0 | 0 | 71 |
| Adoptive relatives of 24 control adoptees | 1 (1.8) | 0 | 0 | 0 | 55 |

*One-tailed P=.0321 for biological index vs control (Fisher's Exact Test).
†One-tailed P=.0129 for biological index vs control (Fisher's Exact Test).

**Table 5. Provincial Sample: Affective, Other Psychotic, and Any Other Diagnoses in Relatives of Adoptees With Chronic Schizophrenia and in Relatives of Control Adoptees With No History of Major Mental Illness**

| Relative Groups | Major Affective, No. (%) | Other Psychotic, No. (%) | Minor Affective, No. (%) | Any Other, No. (%) | Total No. of Relatives |
|---|---|---|---|---|---|
| Biological relatives of 29 adoptees with chronic schizophrenia | 19 (11.1) | 1 (0.6) | 5 (2.9) | 32 (18.7) | 171 |
| Biological relatives of 24 control adoptees | 12 (9.9) | 0 | 4 (3.3) | 26 (21.5) | 121 |
| Adoptive relatives of 29 adoptees with chronic schizophrenia | 7 (9.9) | 0 | 1 (1.4) | 9 (12.6) | 71 |
| Adoptive relatives of 24 control adoptees | 2 (3.6) | 0 | 2 (3.6) | 10 (18.2) | 55 |

syndromes with features resembling some aspect of schizophrenia.

If schizophrenia were simply the extreme end of a normal and continuous distribution of a set of behaviors that ranged from normalcy through schizoid personality to latent and chronic schizophrenia, one might expect the diagnosis of schizoid personality to be more common than the diagnosis of latent schizophrenia, as there are more individuals near the center of a normal distribution. Such was not the case in the present report. Schizophrenic illness appears to be characterized by the definable and identifiable syndromes of chronic and latent schizophrenia, rather than being a variation of normalcy as may be the case for schizoid personality.

## PREVALENCE OF NONFAMILIAL SCHIZOPHRENIA

In this study, as in the Copenhagen Study, very close to half (14 of 29) of the probands with chronic schizophrenia had one or more instances of schizophrenia or latent schizophrenia among their biological relatives, consistent with the operation of genetic factors in the transmission of schizophrenic illness. The remaining 15 index probands whose biological families appeared to be free of those disorders could exemplify a nongenetic, nonfamilial form of schizophrenia or forms in which nongenetic influences predominate. However, the relatively low risk for schizophrenia in the biological relatives of schizophrenic individuals alternatively suggests that these unaffected families of schizophrenic probands were the consequence of small family size and low familial risk for illness rather than evidence of nonfamilial schizophrenia.

Specifically, in the present study we observed a mean family size of 6.0 biological relatives per index proband, and found the risk of either chronic or latent schizophrenia in a biological relative to be 12.9%, similar to the values found in other studies of familial risk.[23(p96)] Given such an observed family size and risk, one would predict that by chance 44% of the families would be unaffected even if all probands had a familial form of the disorder (calculated via the binomial probability distribution).[24(pp126-129)] We observed that 52% of the families appeared to be unaffected, a difference from expectation that is not significant. Thus, the observed distribution of schizophrenia and latent schizophrenia in the families of the probands is quite compatible with an inference that all cases were familial. The alternative hypothesis, that the risk of chronic or latent schizophrenia among the relatives of probands with familial chronic schizophrenia is on the order of 24%, while absent among the probands with putatively nonfamilial schizophrenia (accounting for the observation of illness in only about one half of our families), would require that consistency in familial risk for schizophrenia observed in different samples be accounted for by stability in the ratio of familial to nonfamilial probands.

## THE SEPARATION OF GENETIC AND ENVIRONMENTAL FACTORS

There are possible biases and sources of error to be considered in evaluating the results of this study. These are

**Table 6. Copenhagen Sample: Diagnoses in Relatives of Adoptees With Chronic Schizophrenia and in Relatives of Control Adoptees With No History of Major Mental Illness**

| Relative Groups | Paranoid, No. (%) | Schizoid, No. (%) | Latent Schizophrenia, No. (%)* | Chronic Schizophrenia, No. (%)† | Total No. of Relatives |
|---|---|---|---|---|---|
| Biological relatives of 18 adoptees with chronic schizophrenia | 1 (0.9) | 9 (8.3) | 16 (14.8) | 6 (5.6) | 108 |
| Biological relatives of 23 control adoptees | 1 (0.9) | 15 (13.3) | 1 (0.9) | 1 (0.9) | 113 |
| Adoptive relatives of 18 adoptees with chronic schizophrenia | 0 | 1 (2.5) | 2 (5.0) | 0 | 40 |
| Adoptive relatives of 23 control adoptees | 0 | 4 (6.5) | 2 (3.2) | 0 | 62 |

*One-tailed P=.00006 for biological index vs control (Fisher's Exact Test).
†One-tailed P=.0526 for biological index vs control (Fisher's Exact Test).

**Table 7. National Sample: Diagnoses in Relatives of Adoptees With Chronic Schizophrenia and in Relatives of Control Adoptees With No History of Major Mental Illness**

| Relative Groups | Paranoid, No. (%) | Schizoid, No. (%) | Latent Schizophrenia, No. (%)* | Chronic Schizophrenia, No. (%)† | Total No. of Relatives |
|---|---|---|---|---|---|
| Biological relatives of 47 adoptees with chronic schizophrenia | 4 (1.4) | 16 (5.7) | 30 (10.8) | 14 (5.0) | 279 |
| Biological relatives of 47 control adoptees | 2 (0.9) | 19 (8.1) | 4 (1.7) | 1 (0.4) | 234 |
| Adoptive relatives of 47 adoptees with chronic schizophrenia | 0 | 1 (0.9) | 2 (1.8) | 0 | 111 |
| Adoptive relatives of 47 control adoptees | 1 (0.9) | 4 (3.4) | 2 (1.7) | 0 | 117 |

*One-tailed P=.00002 for biological index vs control (Fisher's Exact Test).
†One-tailed P=.0013 for biological index vs control (Fisher's Exact Test).

essentially the same as those pertinent to the Copenhagen Study, which were discussed in our original reports of that study[1,2] and in our response[25-27] to questions raised by others. The research design of these studies minimizes subjective and selective bias in ascertainment and diagnosis, and independent diagnosis by another group using widely accepted operational criteria[16] has essentially confirmed the findings. The problem of selective placement that is often cited in connection with adoption studies is not as applicable to these for a number of reasons, many peculiar to the Danish adoption sample.

The adoption strategy depends on the ability of that approach to dissociate and randomize the genetic and family-dependent rearing variables in the transmission of the disorders studied. There are circumstances, however, that could make that dissociation imperfect. One circumstance that could operate to diminish the complete separation or randomization of genetic and environmental variables is the possibility of selective placement. Since the etiological role of environmental variables remains obscure at present, it is not likely that a social agency, even if it set about doing so deliberately, could find sufficient of the unknown variables in the prospective adoptive parents to materially affect the risk of illness in the adoptee. One environmental variable, however, that appears to be correlated with the prevalence of schizophrenia is socioeconomic status (SES), although conclusive evidence that it plays an etiological role is lacking. A study of this question, carried out in the Copenhagen adoption sample, found that adoptees who came from low SES biological parents or were placed in low SES adoptive homes were no more likely to develop schizophrenia than adoptees from high SES biological parents or those placed with high SES adoptive homes. Schizophrenic adoptees, however, tended to move into lower socioeconomic strata.[28] The correlation between SES of biological and adoptive parents provides some measure of the degree of selective placement based on that measure. Where such correlations have been measured in Denmark and elsewhere, they have been low. In one study of 206 adoptions in Colorado,[29] average weighted correlations between biological and adoptive parents for education and occupation were .19 and .13, respectively. Teasdale[30] found a correlation coefficient of .15 between SES of biological and adoptive fathers in 11 094 adoptions in the national sample in Denmark, where both data were available. These correlations, though statistically significant, are both small and indicate that selective placement on the basis of SES would not account for more than 3% of the variance. In neither the Copenhagen nor the Provincial Sample was there a significant correlation between socioeconomic class of biological and adoptive parents. Furthermore, in these samples of adoptees, selective placement on the basis of mental illness in the parents could hardly have occurred, since social and legal processes were operating that would select biological and adoptive parents quite free of mental illness. Prospective adoptive parents were screened for mental health and economic stability by the courts and a history or current indication of mental illness in either biological parent would favor a legal abortion or a decision not to place the child for adoption.

The Mothers' Aid Organization, a state social agency that counseled the biological mother and arranged for adoption or other outcome was the repository of information its staff had gathered on the mental status and psychiatric history of the biological and adoptive parents. As a result of the restrictive policies regarding adoption, only two of the 84 initial probands had a biological parent with a history of admission to a mental hospital in the records of the Mothers' Aid Organization at the time of adoption. In neither of these instances, one each among the index and control probands, was the selection of the adoptive parents made by that organization nor did the parental history of mental disturbance play a role in the placement.

Thus, mental illness found in the biological and adoptive parents developed, in practically every case, well after the placement of the adoptee took place. Subsequent search of the Psychiatric Register for dates of all mental hospital admissions of relatives indicated that the average interval between the date of birth of the adoptee and the date of first psychiatric admission for the relatively few biological parents ever hospitalized was 15.5 years and 11.3 years for the Provincial and Copenhagen studies, respectively. A major advantage of the means of ascertainment used in these studies, from a total sample of adoptees rather than from the adopted-away offspring of schizophrenic mothers, was to minimize the effect of schizophrenia in a biological parent, on placement, rearing, personality development, and the recognition of mental illness or its diagnosis in the adoptee. It should also be pointed out that these factors as well as the possibility of nonrandom or selective placement, pertinent, if not adequately controlled, to the development and diagnosis of schizophrenia in adoptees, will have little effect on the prevalence of schizophrenia in the relatives of schizophrenics, which it is the aim of family studies such as this one to ascertain.

More important than selective placement is the possibility of knowledge on the part of adoptive parents, adoptees, and mental health personnel regarding the presence of mental illness in the biological parents and the effect such knowledge could have on the occurrence, perception, and diagnosis of mental illness in the adoptee. Although this problem could be of some significance in studies where the adoptee sample represented children born of mentally ill mothers, it would be negligible in the Danish adoptee sample for reasons that have been discussed.

In contrast to the potential problems in separating genetic and environmental influences that have been discussed above, and that, in Denmark, are minimized, there is the period of time before separation in which the child shared the environment of the biological parents. Ideally this should have been as short as possible, but because of the expected small size of the sample we were reluctant to limit the sample to those adoptees separated immediately after birth. There were, however, more than 80% of the index and control adoptees who were separated from their biological mothers during the first month, 60% within the first week. Restricting the analysis to the 22 index and the 20 control adoptees who spent no more than 4 weeks with their biological families, we find eight of the 128 biological index relatives who developed schizophrenia and none of the 105 biological control relatives (Fisher's Exact Test, one-tailed $P=.0075$). A similar pattern is observed for latent schizophrenia (eight of 128 index relatives vs three of 105 control relatives), which is not statistically significant. These results, similar to those found in the Copenhagen Study, indicate that the striking concentration of schizophrenia found in the biological relatives of schizophrenic adoptees is not accounted for by the period of time spent with the biological mother or parents. All of the adoptees in the present study who were found to have schizophrenia in their biological relatives had stayed with the natural mother for 2 weeks or less.

However, every adopted child has spent 9 months in the uterus of its biological mother, where it shares with her a number of environmental influences, and, in most cases, has had a certain amount of early mothering at her hands. It is possible to disentangle that conjunction of genes and environment by examining the prevalence of the disorder in the biological paternal half-siblings of the adoptees. These shared neither the same uterus nor the same early mothering with the adoptee, but a certain amount of genetic overlap. In both the Copenhagen and the Provincial Samples, the prevalence of chronic or latent schizophrenia in the paternal biological half-siblings was at least as high as in the maternal half-siblings, and significantly higher than in their controls.

## DIAGNOSTIC VALUE OF GLOBAL CRITERIA COMPARED WITH THOSE BASED ON *DSM-III*

A serious deficiency in psychiatric diagnosis is its dependence on current usage in the absence of independent criteria for validity. Pathognomonic features and laboratory findings of proved specificity have served as valuable support for the structure of diagnostic systems in medicine and much of the confusion that has accompanied the diagnosis and study of schizophrenia would have been obviated had there been a serological, chemical, psychological, or clinical test with acceptable sensitivity and specificity. The development of *DSM-III* and its successors, operational diagnostic systems endorsed by the American Psychiatric Association, has improved the interrater reliability and communicability of many psychiatric diagnoses, as was to be expected. The question of validity, however, has not often been addressed or adequately answered.

The prevalence of schizophrenia in the biological relatives of schizophrenic adoptees and its relative absence in normal control relatives can be a test not only of hypotheses of genetic transmission, but also of the sensitivity and specificity with which that syndrome is detected by the diagnostic criteria chosen. Diagnoses in the Copenhagen Study were global judgments based on the descriptions in *DSM-II*, the diagnostic manual then in use. For schizophrenia the original and more complete descriptions of Kraepelin[15] and Bleuler[9] were employed. It was important to use the same global diagnostic system in the Provincial Study if it was to be a replication of the Copenhagen Study in a new Danish sample.

Despite the fact that *DSM-III* became available about the time that the diagnoses on the relatives in the present study were to be made, it did not supplant the global diagnoses in this study for several additional reasons. The focus of *DSM-III* on a minimum number of putative characteristics made it useful for quick and economic diagnosis, but its necessary neglect of a large number of potentially important features diminished its sensitivity and usefulness in research. Interrater reliability would be enhanced, as expected, for any system using criteria agreed on in advance by the raters, and our diagnostic system had criteria previously agreed on by the raters and with comparable interrater reliability. Furthermore, the validity of the *DSM-III* diagnosis of schizophrenia and cognate diagnoses, in common with that of other operational diagnoses, had not been tested independently. When Kendler and Gruenberg[16] asked to review the interviews from our Copenhaqen sample to determine whether a genetic relationship between schizophrenia and anxiety neurosis was observable, an opportunity became available for independent diagnoses of schizophrenic illness using *DSM-III* and a comparison with global diagnoses. They were encouraged to extend their diagnostic ratings into the schizophrenia spectrum as well.

The Copenhagen adoption study provided a means of evaluating the validity of the global and *DSM-III* diagnoses of schizophrenia and related syndromes and comparing them for sensitivity and specificity. The *DSM-III* diagnoses made independently by Kendler and Gruenberg[16] from the interviews alone in the Copenhagen Study appeared to confirm but did not improve the results of our global diagnoses on the same interviews,[2] which encouraged us to continue to use global diagnoses in the present study.

## CONCLUSION

This communication reports the results of the second phase of a national study, replicating the design of the first phase that was confined to the city and county of Copenhagen and offers the advantage of testing hypotheses examined or derived in the first study in an additional and independent sample. The design of both studies involves matched case-controls, minimizes ascertainment and subjective bias, and is unaffected by selective placement. Results from these two major divisions of the Danish adoption sample at the level of the family and at the level of individual family members, each involving specific diagnoses, support the presence of a strong genetic liability in a major segment of schizophrenia.

Since no excess of such liability has been observed among natural families when compared with the adoption samples we have investigated, the familial tendency in schizophrenia appears to be largely an expression of genetic factors, lending important justification to the use of natural pedigrees for investigating the genetics of schizophrenia. The observation that in both population samples a syndrome based on the classic descriptions of Kraepelin[15] and Bleuler[9] is found in certain adoptees and in their biological relatives consistently and almost exclusively is evidence for the reliability and validity of their concept of schizophrenia as well as its genetic transmission.

Investigation of the sensitivity and specificity of diagnostic criteria to identify illness in the biological families of adoptees has the potential to provide an empirical basis for diagnosis. Furthermore, such studies provide an approach for assessing the validity of psychiatric diagnoses in reflecting the operation of genetically determined underlying biological mechanisms. If comparison between our global diagnoses and *DSM-III* diagnoses in the Provincial Sample mirrors the results of the previous comparison of diagnostic systems in the Copenhagen Study, this will provide further indication that global diagnoses are capable of greater sensitivity and equal specificity when compared with operational diagnoses exemplified by *DSM-III*. In both the Copenhagen and Provincial studies, a milder syndrome, latent schizophrenia, was found significantly concentrated in the biological relatives of schizophrenic adoptees at an even greater prevalence than classical schizophrenia. This speaks for a genetic relationship between the two syndromes. The lack of specificity of latent schizophrenia to the biological relatives of individuals with chronic schizophrenia suggests heterogeneity of latent schizophrenia as presently defined.

The results suggest that inconsistencies in the familial distribution of schizophrenia due to nongenetic, sporadic schizophrenia are probably of minor importance. Nevertheless, clarifying the potential existence (and origin) of a putative nongenetic form of schizophrenia remains a valid goal.

The demonstration of a genetic liability for schizophrenia in no way diminishes the role that the environment plays in the initiation, course, and outcome of schizophrenic illness and is not incompatible with external physical and chemical, or viral, autoimmune, and developmental etiological influences and processes. Continued investigation of the genetic and extragenetic contributions will be necessary to elucidate the modes of genetic transmission and expression, and to identify and define the large number of environmental influences operating in tandem with them in the development of this disorder.

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